

Effect of local infiltration with oxytocin on hemodynamic response to surgical incision and postoperative pain in patients having open laparoscopic surgery under general anesthesia

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Abstract

Background: Preemptive analgesia encompasses different perioperative interventions that have the final aim of decreasing postoperative pain and improving recovery. Recently, peripheral analgesic effects of oxytocinergic modulation have been suggested. In this regard, we tested the potential analgesic effects of subcutaneous oxytocin (OT) infiltration in patients submitted to laparoscopic cholecystectomy.

Methods: Thirty patients with similar general characteristics and medical physical conditions were evaluated. The patients were assigned by simple random selection to one of three groups: (a) OT group ($n = 10$), which received preincisional subcutaneous OT (4 μ g/4 ml saline) in the surgical sites for trocar placements; (b) Lidocaine group ($n = 10$), which received subcutaneous lidocaine 1% (4 ml) in the surgical sites; and (c) Control group ($n = 10$), which did not receive any treatment. Then we measured the effect of those treatments on the hemodynamic variations produced as responses to the surgical incisions and trocar insertions (open port placements using the Hasson technique). Moreover, we assessed the intensity of postoperative pain with the visual analogue scale during recovery and 24 hr after surgery.

Results: Hemodynamic parameters were stable in both intervention groups (subcutaneous OT and lidocaine) during the surgical incisions and trocar placements, whereas a most likely sympathetic activation due to trocar insertions (open port placements) was not blunted in the placebo group. Furthermore, postoperative pain was diminished in both OT and lidocaine groups when compared to the control group.

Conclusions: Preincisional subcutaneous OT administration reduced the hemodynamic response produced by the latter. Furthermore, OT also diminished postoperative pain.

1 | INTRODUCTION

Postoperative pain is a major clinical problem, and its treatment and prevention continue to be a challenge to clinicians. Surgical intervention can produce an entity named persistent postoperative pain (PPP) which is characterized by chronic pain that continues after surgical lesions have healed, affects 2%–10% of postsurgical patients, and considerably reduces their quality of life and surgery recovery (Cregg, Anwar, & Farquhar-Smith, 2013; Werner and Kongsgaard, 2014). PPP is considered a growing health issue with a large population at risk (Macrae, 2008) and high prevalence (Cregg et al., 2013).

One measure that has been recently under intense development is preventive analgesia (PA). PA encompasses different perioperative interventions that have the final aim of decreasing postoperative pain and the consumption of analgesics (Vadivelu et al., 2014; White, 2008; Woolf & Chong, 1993). Besides, it has been shown that PA reduces the incidence of PPP (Lavand'homme, De Kock, & Waterloo, 2005). Different techniques of PA have been studied, including neuraxial, systemic and local infiltration of several drugs (Vadivelu et al., 2014).

Local infiltration in the surgical area before incision using anesthetic agents is already a common practice that has proven to be useful to reduce the intensity of postoperative pain, probably through a deafferentation process (Ejlertsen, Andersen, Eliassen, & Mogensen, 1992). Nevertheless, the most common agents (i.e., bupivacaine), especially at large doses, can produce adverse effects (Kakagia, Fotiadis, Tripsiannis, & Tsoutsos, 2007). Also, it is considered that the effect of the currently used drugs is limited by their duration of action (White, 2008). Therefore, the inclusion of adjuvants or drugs with different mechanisms of action seems suitable to improve the PA.

Oxytocin (OT) is an interesting molecule that has demonstrated analgesic properties in basic and clinical research. Central administration of OT produces antinociception in rodents (Condés-Lara, Martínez-Lorenzana, Rodríguez-Jiménez, & Rojas-Piloni, 2008; Condés-Lara, Martínez-Lorenzana, Rojas-Piloni, & Rodríguez-Jiménez, 2007; Condés-Lara, Rojas-Piloni, Martínez-Lorenzana, & Rodríguez-Jiménez, 2009; Condés-Lara et al., 2006) and potent analgesia in humans (Condés-Lara et al., 2016) without adverse effects. Recently it was shown, using electrophysiology and behavioural pain models, that local peripheral (subcutaneous) administration of OT produces long-lasting antinociception in rats (González-Hernández et al., 2017). Using pharmacological and histologic approaches, González-Hernández et al., showed that the oxytocinergic antinociceptive effect was achieved through the local activation of OT receptors expressed in peripheral

Highlights

- Preincisional subcutaneous oxytocin reduces nociceptive response to surgical incision.
- Subcutaneous oxytocin reduces postoperative pain.

nerve terminals in the skin. Immunofluorescence revealed that these OT receptors are co-expressed with the pronociceptive peptide CGRP but not with the IB4 marker. These findings were the rationale to assess if subcutaneous OT administration also produces antinociception and analgesia in patients subjected to surgery.

The primary objective of the present study was to evaluate the antinociceptive and analgesic effect of subcutaneous preincisional OT infiltration in the surgical area for open port placements (performed with the Hasson technique) in patients subjected to laparoscopic cholecystectomies. Nociception was assessed by comparing the hemodynamic response (autonomic nociceptive response) produced by surgical incisions between patients infiltrated with OT vs. preincisional lidocaine or no treatment (control).

Then we measure the analgesic component of OT treatment with the quantification of immediate and short-term postoperative pain using the visual analogue scale (VAS).

2 | MATERIAL AND METHODS

2.1 | Subjects studied

Our institutional bioethics committee approved the study and it was conducted following the Good Clinical Practice guidelines and principles expressed in the Helsinki Declaration. All patients received a careful explanation of the study, and they gave their informed consent. The trial registration number is ISRCTN29008609.

The protocol was performed in patients who underwent scheduled laparoscopic cholecystectomy and gave their participation informed consent. We included adult patients with an American Society of Anesthesiologists (ASA) Physical Status Class I and II (e.g., healthy participants or participants with compensated systemic disease). We excluded from the protocol patients with ASA Class II and systemic hypertension, in treatment with antihypertensive medications, ASA Class III or more (decompensated systemic disease), hemodynamic instability (arterial pressure below 80/40 or bradycardia), patients with hypersensitivity to any medication used in the study, patients with chronic use of opioids or neuromodulatory drugs, and patients with acute pain due to cholecystitis. We also, excluded patients

TABLE 1 General characteristics and clinical variables. The three groups showed no difference in age, body mass index, ASA scale score, anesthetic consumption, duration of surgery, liquid balance (input/output) (data are presented in average $\pm$ standard error; comparison was performed with a one-way ANOVA with Tukey post hoc test)

	Control	OT	Lidocaine	<i>p</i> -value
General variables				
Age (years)	46.6 $\pm$ 4.3	46.9 $\pm$ 3.3	42 $\pm$ 2.9	0.553
Body mass index	26.3 $\pm$ 1.8	28.1 $\pm$ 1.5	27.5 $\pm$ 0.9	0.719
Gender (male/female)	2/8	3/7	3/7	–
Score, American Society of Anesthesiologists scale				
Asa I	3	4	6	–
Asa II	7	6	4	–
Anesthetic consumption (mg/patient)				
Midazolam	3.3 $\pm$ 0.17	3.45 $\pm$ 0.1	3.550 $\pm$ 0.2	0.751
Fentanyl	309 $\pm$ 21.8	286.7 $\pm$ 16.4	293.2 $\pm$ 12.2	0.641
Rocuronium	43.4 $\pm$ 1.5	45.4 $\pm$ 1.2	44.1 $\pm$ 1.1	0.548
Duration (min)				
Pneumoperitoneum time	103.2 $\pm$ 3.7	107.5 $\pm$ 5.4	105.5 $\pm$ 5.2	0.825
Surgery time	109.8 $\pm$ 3	112.2 $\pm$ 5.6	110 $\pm$ 5.4	0.899
Anesthetic time	120.8 $\pm$ 2.8	120.8 $\pm$ 2.8	126.1 $\pm$ 6.2	0.718
Input (ml)				
Crystalloid input	1700 $\pm$ 44.8	1680 $\pm$ 66.7	1590 $\pm$ 42.8	0.315
Output (ml)				
Estimated blood (loss)	ns	ns	ns	–
Urine output	205.6 $\pm$ 4.12	189.4 $\pm$ 16.6	195 $\pm$ 16.5	0.707
Peak airway pressure	17.8 $\pm$ 1.2	18.3 $\pm$ 1.08	18.2 $\pm$ 2	0.245

under treatment with antihypertensive medications or under chronic analgesic treatment (with opioids and neuro-modulatory drugs). None of our patients received inotropic drugs. Elimination criteria included complications during surgery, e.g., bleeding (more than 500 cc), conversion to open surgery due to technical difficulties, intestinal injury and use of inotropic drugs (ephedrine, atropine) during surgery.

A total of 32 adult volunteers of both sexes participated in the study, our population had an average age of 45.16 $\pm$ 12 years (*SE*) (control group: 46.6 $\pm$ 4.3; OT group: 47 $\pm$ 3.3; lidocaine group: 42 $\pm$ 2.9; *p* = 0.553) and a Body Mass Index (BMI) of 27.3 $\pm$ 5 (control group: 26.3 $\pm$ 1.8; OT group: 28.1 $\pm$ 1.5; lidocaine group: 27.5 $\pm$ 0.9; *p* = 0.719). 43.3% of patients were classified as ASA I, and 56.6% of cases had an ASA II. No statistical differences in age, physical characteristics and general conditions were found between groups (Table 1).

Of the 32 initial volunteers, 1 patient in the OT group was eliminated due to surgical complications (bleeding more than 500 cc during gallbladder dissection, with conversion to open surgery), and 1 patient in the control group was eliminated due to hypertension after surgery (unconventional postoperative evolution with arterial blood pressure up to 160/95). Thirty patients concluded the protocol (enrolment diagram Fig. 1).

2.2 | Surgical procedure and drug administration

Standard general anesthesia was used in all patients. We used midazolam 0.05 mg/kg, fentanyl 0.004 mg/kg, propofol 2 mg/kg and rocuronium 0.6 mg/kg for the anesthetic induction. Facemask ventilation in the interval between induction and intubation. Then, orotracheal intubation 2 min after administration of the neuromuscular blocking agent, and initiation of controlled mechanical ventilation, with tidal volume 8–10 ml/kg. Anesthesia maintenance was performed using oxygen (97%) and sevoflurane (2.5% end-tidal fraction at 1.5 minimum alveolar concentrations). Patients' ventilation and respiratory parameters were controlled to maintain an ET CO₂ between 35 and 40 mmHg after the installation of the capnoperitoneum; this was performed in order to compensate for the possible peritoneal CO₂ absorption.

The patients were assigned by simple random selection to one of three groups: (a) OT group (*n* = 10) who received subcutaneous OT (Oxitopisa® 4 µg in 4 ml of saline) in each of the surgical sites before the incisions for the placements of three or four trocars. (b) Lidocaine group (*n* = 10) who received subcutaneous infiltration of lidocaine 1% (Pisacaina 1% ® 4 ml). (c) Control group (*n* = 10) who did not receive any treatment at the surgical site. The patients and the surgeon were blinded to the composition of the injected solution. The

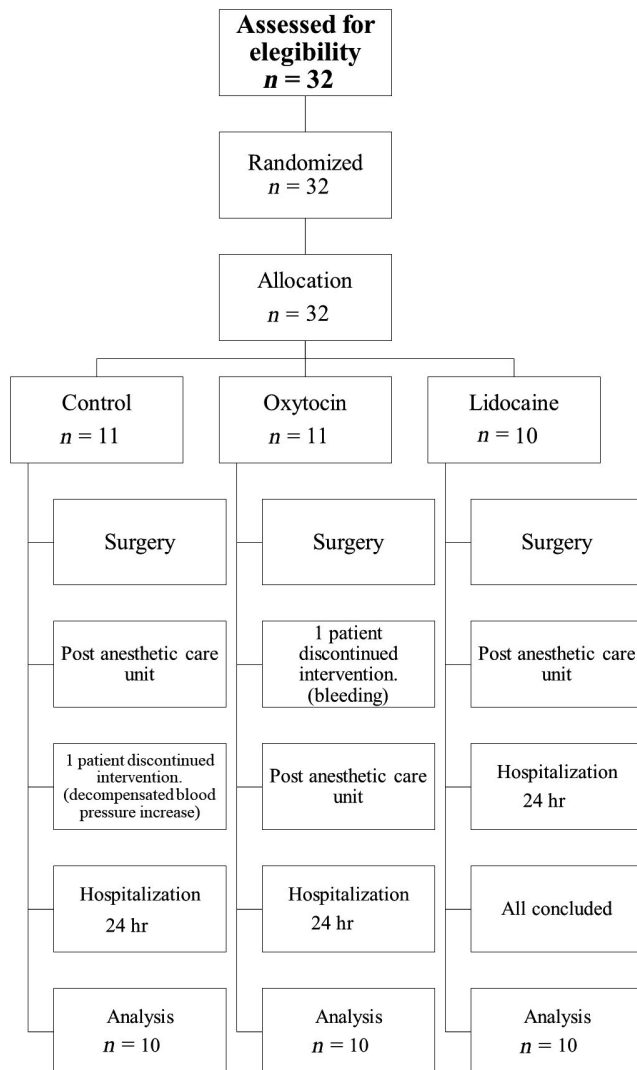


FIGURE 1 Flow diagram. A total of 32 patients were selected, 1 patient of the OT group was eliminated due to surgical bleeding (specified in the text), 1 patient of the control group was eliminated due to decompensated blood pressure increase (specified in the text) and 30 patients culminated the protocol

surgical procedure used to insert the trocars was performed with the Hasson technique.

In the three groups, the following parameters were recorded with a GE Health Care model Datex Ohmeda S5 multiparameter compact anesthesia monitor™ during the operation: heart rate, blood pressure (systolic, diastolic and mean), pulse oximetry. Also, ventilatory parameters were measured during all the procedure using a General Electric Datex Ohmeda Anesthesia carestation S/5 Avance Monitor™, including end-tidal CO₂, tidal volume, respiratory rate, airway pressure. Although these parameters were recorded during the entire procedure, they were documented specifically at the following time points: Basal measurement, anesthetic induction, drug infiltration (corresponding to the post-intubation) during the surgical incisions and trocar

placements (we took the maximum value presented from the incision until the end of the trocar placements), transurgical (insufflation of pneumoperitoneum and the maximum value presented during the gallbladder dissection) and anesthetic emersion (signs after extubation before admission to the post-anesthetic care unit, PACU). Additionally, the requirement of intraoperative salvage administration of fentanyl (0.001 mg/kg) as a response to an increment of blood pressure (>20%) was also documented.

General postoperative pain at rest was assessed using the visual analogue scale (VAS, 0–10) after the patient's admission to and discharge from the PACU and during hospitalization 24 hr after surgery. During their stay in the PACU, patients received morphine (0.05 mg/kg) as a rescue treatment. Then, patients were given a postoperative analgesic scheme with scheduled ketorolac 60 mg/8 hr and, if VAS was greater than 5, salvage treatment with tramadol 50 mg. The number of salvage doses was recorded for postoperative pain evaluation. Finally, the occurrence of adverse effects was also assessed.

2.3 | Statistical analysis

Statistical analysis was performed using GraphPad Prism V6.01 software (La Jolla, CA). The Shapiro–Wilk test was used to evaluate the normal distribution of data. A one-way repeated measures analysis of variance (one-way ANOVA) with a Tukey post hoc test were used to compare the changes in cardiovascular parameters (heart rate and systolic and diastolic pressure) evoked by surgical incisions. Postoperative pain values (VAS) between groups were compared using the non-parametric Kruskal–Wallis test, with a Tukey post hoc test. The data are expressed as mean, standard deviation, and 95th and 5th percentiles. Statistical significance was considered with p -values < 0.05.

3 | RESULTS

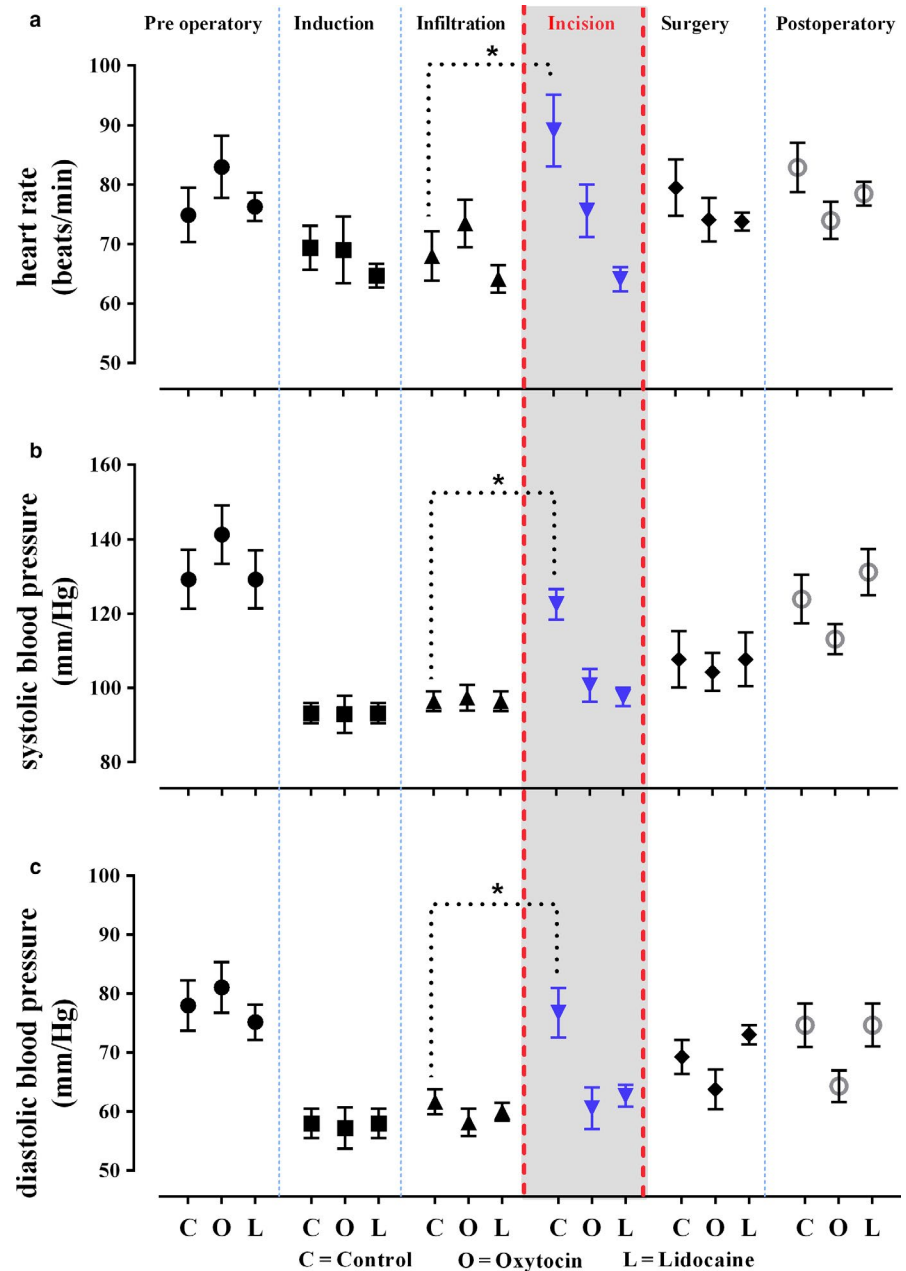
3.1 | General variables

The general measures of the participants showed no significant statistical difference between the three groups (control, lidocaine and oxytocin) in the following general variables: (a) ASA scale, (b) age, (c) body mass index, (d) gender, (e) anesthetic consumption, (f) surgery duration, (g) fluid balance and (h) peak airway pressure (Table 1).

3.2 | Cardiovascular parameters

The nociceptive autonomic response (increased heart rate and diastolic and systolic blood pressure) to surgical incisions and trocar insertions was blunted by the subcutaneous

FIGURE 2 Cardiovascular parameter variations. The cardiovascular parameters (a) heart rate, (b) systolic blood pressure and (c) diastolic blood pressure are depicted throughout the different surgery phases. Each point corresponds to the mean value with standard errors. Vertical dashed lines separate each surgery phase. Grey area delimited by red dashed lines indicate the pharmacological oxytocin (O) or lidocaine (L) effect. Notice the blunt cardiovascular response to incision produced by both treatments. * $p < 0.05$, one-way ANOVA, Holm–Sidak post hoc test



preincisional infiltration of OT and lidocaine (Fig. 2). In particular, heart rate mean values in the control group increased by 14% (21 ± 3.4 beats per minute (bpm); $p = 0.001$), compared to 2.8% (2.01 ± 1.6 bpm; $p = 0.331$) in the OT group and 1.61% (1.1 ± 0.69 bpm; $p = 0.646$) in the lidocaine group. Moreover, the increase in systolic blood pressure was of 27% (26.10 ± 4.7 mmHg; $p = 0.012$) in the control group, compared to 3.3% (3.3 ± 3.9 mmHg; $p = 0.345$) in the OT group and 2.4% (2.4 ± 0.89 mmHg; $p = 0.24$) in the lidocaine group (Fig. 2). Finally, the increase in diastolic blood pressure was of 24% (15.1 ± 3.5 mmHg; $p = 0.006$) in the control group, compared to 4.1% (2.4 ± 3.5 mmHg; $p = 0.280$) in the OT group and 2.7% (1.7 ± 0.77 mmHg; $p = 0.6039$) in the lidocaine group (Fig. 2).

3.3 | Salvage treatment

The transoperative salvage fentanyl dose was 0.001 mg/kg. During surgery, four patients of the control group received a total of four fentanyl rescue administrations, one per patient. In the OT group, only one patient needed a fentanyl rescue, and in the lidocaine group, two patients required fentanyl rescues.

3.4 | Respiratory parameters

There were no significant differences between the groups in pulse oximetry and end-tidal CO_2 during the procedure (data not shown). The control group had a peak airway pressure of

17.8 ± 1.2 cmH₂O, whereas the lidocaine and OT groups had values of 18.3 ± 1.08 cmH₂O and 18.2 ± 2 cmH₂O, respectively ($p = 0.245$) (Table 1).

3.5 | Postsurgical pain

The mean VAS scores were similar between groups upon the patients' arrival in the PACU. The control group had a mean VAS score of 2.6 ± 0.3, the OT group of 1.7 ± 0.2 ($p = 0.093$) and the lidocaine group of 1.9 ± 0.1 ($p = 0.076$) (Fig. 3). However, a statistically significant difference was found between the control group and intervention groups upon discharge from the PACU. The control group had a mean VAS score of 3.8 ± 0.53, the OT group of 2.1 ± 0.23 ($p = 0.001$) and the lidocaine group of 2.2 ± 0.35 ($p = 0.007$) (Fig. 3). Finally, 24 hr after surgery, the mean VAS scores were 3.6 ± 0.45 for the control group, 1.7 ± 0.39 for the OT group ($p = 0.0038$) and 2.1 ± 0.27 for the lidocaine group ($p = 0.07$).

3.6 | Postoperative pain management

Morphine (0.05 mg/kg) was administered for postoperative treatment in the PACU. Five patients in the control group received morphine rescues, whereas while only one patient in the OT group and one patient in the lidocaine group received morphine rescues. Finally, during hospitalization only one patient in the control group received tramadol. We only describe salvage administrations because hardly any patients required them.

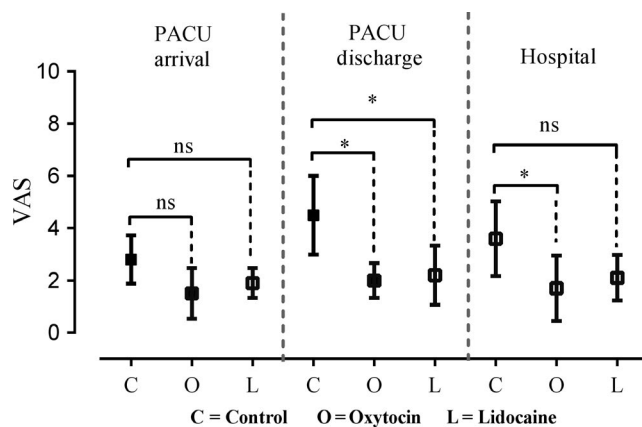


FIGURE 3 Postsurgical pain. Pain measured by the visual analogue scale

Note: Pain measured by visual analogue scale (VAS) mean values is represented at the arrival to the post-anesthesia care unit (PACU), at the discharge from the PACU and during hospitalization (24 hr after surgery). Data are depicted as mean with maximum and minimum values. Notice the statistical decrease in VAS values after oxytocin (O) treatment. * $p < 0.05$, Kruskal–Wallis non-parametric test, Tukey's post hoc test.

3.7 | Adverse effects

At discharge, the surgeon examined the surgical wounds, which were unremarkable in all three groups. None of the patients manifested local or systemic adverse effects.

4 | DISCUSSION

The measurement of cardiovascular parameters during the anesthetic procedure gives useful information on different aspects of the anesthetized patient. Heart rate and blood pressure quantification are important for measuring the trans-surgical nociceptive process and its response to treatment (Guignard, 2006). Recently, it has been suggested that the diminution of the cardiovascular response to nociceptive stimuli is relevant not only for neurovegetative protection but also for the appearance of postoperative pain. Poor control over heart rate changes associated with the surgical stimulus increases the risk of maintaining high scores of pain in the first hours after surgery (Uchida et al., 2017). Therefore, for many years it has been an extended practice to combine drugs and administer them in several sites with the aim of improving antinociception and reducing the excessive need for analgesics and anesthetics. Inhaled and intravenous drugs (Ogawa et al., 2006), as well as neuraxial (Katz et al., 1992) and infiltration (Barreveld et al., 2013) techniques have been studied.

Preincisional infiltration of different drugs has also shown benefit in postoperative pain and currently is a regular practice in the anesthetic practice. The subcutaneous infiltration of simple local anesthetics has shown efficiency in postoperative analgesia in different studies (Barreveld et al., 2013; Kundra et al., 2016; Mohamed et al., 2018). In addition of reducing postoperative symptoms, the preincisional infiltration of drugs have the aim of diminishing the doses of high-risk medications (e.g., opioids) and their corresponding adverse effects. The latter include respiratory depression, nausea, vomiting (Apfelbaum, Chen, Mehta, & Gan, 2003; Beverly, Kaye, Ljungqvist, & Urman, 2017; Chester, Stanford, & Gazet, 1990; Daniel, Weiskopf, Noorani, & Eger, 1998; Niraj et al., 2011). This is especially true with opioids used in perioperative and postoperative pain management, so much that new directions were recently published to reduce the indiscriminate prescription of this kind of drugs (Dowell, Haegerich, & Chou, 2016). In this sense, the need for new pharmacological approaches with fewer side effects is especially urgent.

Various studies have assessed the analgesic role of OT administration. Cerebroventricular or epidural OT administration produced remarkable analgesia in patients with severe pain and terminal cancer (Condés-Lara et al., 2016; Madrazo, Franco-Bourland, León-Meza, & Mena, 1987). Patients with

low back pain also responded to treatment with intrathecal OT (Yang 1994). The effect of OT intranasal administration in the treatment of different clinical entities such as chronic pelvic pain of somatic origin (Rash et al., 2017) and fibromyalgia (Mameli et al., 2014) was also investigated. However, this is the first clinical report of the analgesic effect of subcutaneous OT. The results from this study are in line with recent basic science reports describing that peripheral activation of OTRs present in skin nerve terminals produces long-lasting antinociception (González-Hernández et al., 2017; Manzano-García et al., 2018; Nersesyan et al., 2017).

In this study, we found that subcutaneous preincisional administration of OT can produce antinociception in anesthetized patients. Preincisional OT blunted the hemodynamic response associated with the nociceptive autonomic reaction elicited by the surgical incisions and trocar placements in patients subjected to laparoscopic cholecystectomy with the Hasson technique. Moreover, during surgery patients infiltrated with OT or lidocaine received fewer fentanyl rescue administrations compared to control patients. Taking all of this into account, our results provide evidence of the antinociceptive effect of OT.

On the other hand, upon arrival in the PACU, both OT and lidocaine-treated patients did not show a significant reduction in the VAS measurements compared to the control group. We hypothesize that this result was conditioned by a residual effect of general anesthesia and the use of transanesthetic fentanyl rescues. However, the preincisional treatment with OT or lidocaine was related to less postoperative pain during discharge from the PACU and during hospitalization compared to control patients. The OT group maintained a statistically significant difference in mean VAS values throughout hospitalization, whereas the lidocaine group did not. Moreover, during the immediate postoperative period the control group required more analgesic rescue treatments with morphine than the patients from the OT or the lidocaine groups.

5 | CONCLUSIONS

Based on the results obtained, we can conclude that OT subcutaneous infiltration reduced the cardiovascular response associated with transoperative nociception compared to the control group. Moreover, patients in the OT group require lower transoperative and postoperative analgesic rescues. Finally, OT treatment was associated with low postoperative pain compared to the control group. Importantly, the effects of OT infiltration were similar to those produced by lidocaine treatment. Studies with larger population cohorts are necessary to advance understanding of the effects of subcutaneous OT. However, preincisional OT has the potential to become a valuable option in perioperative pain management.

ACKNOWLEDGEMENTS

We acknowledge Jessica Gonzalez-Norris for proofreading the manuscript. We thank the National Council of Science & Technology (CONACyT Mexico; Grant No. 164536) and PAPIIT (UNAM Mexico; Grant No. IN200415) to MCL. We also thank the authorities of the Hospital Regional PEMEX Salamanca for supporting the realization of this project.

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CONFLICTS OF INTEREST

The authors declare no conflict of interest. The sponsors had no further role in (1) study design; (2) in the collection, analysis and interpretation of data; (3) in the writing of the report; and (4) in the decision to submit the paper for publication.

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How to cite this article: Zayas-González H, Hernández AG, Manzano-García A, et al. Effect of local infiltration with oxytocin on hemodynamic response to surgical incision and postoperative pain in patients having open laparoscopic surgery under general anesthesia. *Eur J Pain*. 2019;23:1519–1526. <https://doi.org/10.1002/ejp.1427>